



AutoML-Agent Implementation Flowcharts - Text Descriptions

1. System Architecture Overview Flowchart

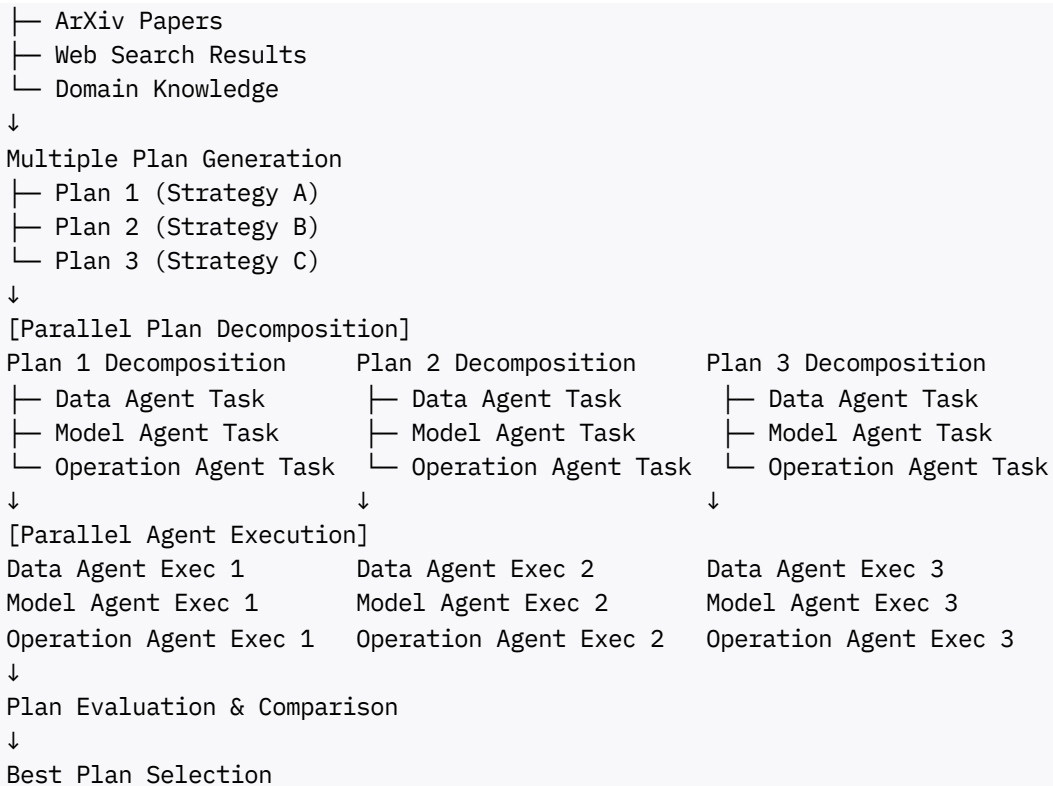
Flow Direction: Top to Bottom with Feedback Loops

```
User Input (Natural Language)
↓
Agent Manager (Central Orchestrator)
↓
Stage 1: Initialization
↓
Request Verification → [Decision Point: Clear & Complete?]
├─ No → Request User Clarification → Loop back to Request Verification
└─ Yes → Continue
↓
Prompt Agent (NL Processing)
↓
Stage 2: Planning
↓
[Parallel Processing]
├─ Data Agent (Data Pipeline)
├─ Model Agent (Model Search)
└─ Operation Agent (Code Generation)
↓
Execution Verification → [Decision Point: Plans Valid?]
├─ No → Automatic Revision → Loop back to Planning
└─ Yes → Continue
↓
Stage 3: Execution
↓
Implementation Verification → [Decision Point: Code Works?]
├─ No → Automatic Revision → Loop back to Planning
└─ Yes → Continue
↓
Deployed Model with API Endpoint
```

2. Retrieval-Augmented Planning (RAP) Process Flowchart

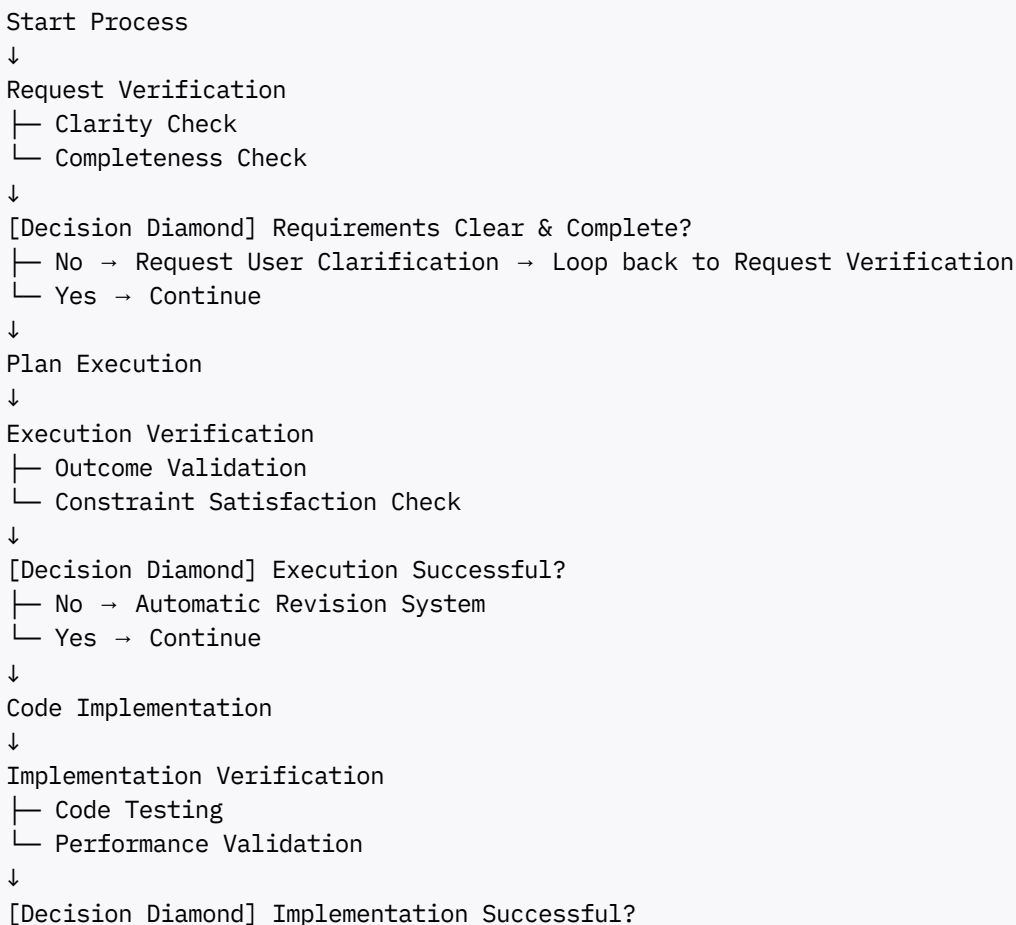
Flow Direction: Left to Right with Parallel Branches

```
User Requirements (Parsed JSON)
↓
External Knowledge Retrieval
```



3. Multi-Stage Verification System Flowchart

Flow Direction: Sequential with Decision Points and Loops



```
├─ No → Automatic Revision System
└─ Yes → Continue
↓
Successful Deployment

[Revision Loop]
Automatic Revision System
↓
Failure Analysis
↓
Plan Improvement
↓
Loop back to Plan Execution
```

4. Detailed Agent Workflow Flowchart

Flow Direction: Sequential with Data Flow Arrows

```
Natural Language Instruction
↓
Agent Manager (Orchestrator)
↓
Prompt Agent
├─ NL Processing & Instruction Parsing
└─ Structured JSON Requirements → Back to Agent Manager
↓
[Parallel Agent Coordination by Agent Manager]

Data Agent Branch:
├─ Dataset Search (Kaggle, HuggingFace)
├─ Data Retrieval & Preprocessing
├─ Statistical Analysis & Profiling
└─ Data Characteristics Summary → To Model Agent & Agent Manager

Model Agent Branch:
├─ Receives Data Summary from Data Agent
├─ Training-Free Model Search
├─ Hyperparameter Optimization Simulation
├─ Model Ranking & Selection
└─ Top-k Model Candidates → To Operation Agent & Agent Manager

Operation Agent Branch:
├─ Receives Top Models from Model Agent
├─ Production Code Generation
├─ Runtime Debugging & Error Handling
├─ API Endpoint Creation
└─ Deployment-Ready Package → To Agent Manager

Agent Manager:
├─ Collects all agent outputs
└─ Final Integration & Deployment
```

5. Enterprise Deployment Architecture Flowchart

Flow Direction: Multi-layered Architecture

```
[User Layer]
User Interface (Web/Mobile/API)
↓
[Gateway Layer]
API Gateway & Load Balancer
↓
[Security Layer]
Security & Access Control (RBAC)
↓
[Application Layer]
AutoML-Agent Core System
├─ Agent Containers (Docker/Kubernetes)
└─ Auto-scaling Infrastructure
↓
[Integration Layer]
External API Integrations
├─ OpenAI GPT-4
├─ HuggingFace Models/Datasets
├─ Kaggle Datasets
└─ ArXiv Papers & Web Search
↓
[Storage Layer]
├─ Model Registry & Versioning
└─ Data Storage (S3/Cloud)
↓
[Monitoring Layer]
├─ Monitoring System (Prometheus/Grafana)
└─ Centralized Logging & Audit Trail
↓
[Deployment Layer]
Deployed Model Endpoints
↓
[Backup Layer]
Backup & Disaster Recovery
```

Key Implementation Steps from Flowcharts:

Phase 1: Core System Setup

1. **Infrastructure Preparation:** Set up containerized environment with Docker/Kubernetes
2. **API Integration:** Configure external API access (OpenAI, HuggingFace, Kaggle, ArXiv)
3. **Agent Development:** Implement five specialized agents with defined interfaces
4. **Verification System:** Build multi-stage verification with automatic revision capability

Phase 2: Agent Orchestration

1. **Agent Manager:** Develop central orchestration logic
2. **Parallel Processing:** Implement concurrent plan generation and execution
3. **Data Flow:** Establish information exchange between agents
4. **Quality Gates:** Integrate verification checkpoints throughout pipeline

Phase 3: Production Deployment

1. **Security Layer:** Implement RBAC and access controls
2. **Monitoring:** Set up real-time performance tracking and alerting
3. **Scaling:** Configure auto-scaling based on demand
4. **Backup/Recovery:** Establish disaster recovery procedures

Phase 4: Enterprise Integration

1. **API Gateway:** Deploy load-balanced API endpoints
2. **Model Registry:** Implement versioning and model management
3. **Audit Trail:** Enable comprehensive logging and compliance
4. **User Interface:** Develop intuitive front-end for business users

These flowcharts demonstrate the sophisticated multi-agent architecture that enables AutoML-Agent to achieve 87.1% success rates while maintaining 8x speed improvements over traditional approaches.